Exercise 11.1 SOLUTION

Q1. Determine whether the following functions are even, odd or neither odd nor even.

i. $\sin^2 x$

Sol: $f(x) = \sin^2 x = (\sin x)^2$

$f(-x) = [\sin(-x)]^2 = (-\sin x)^2 = (\sin x)^2$

$= \sin^2 x = f(x)$

$\Rightarrow f(-x) = f(x) \Rightarrow f(x)$ is an even function

ii. $\sin x + \cos x$

Sol: $f(x) = \sin x + \cos x$

$f(-x) = \sin(-x) + \cos(-x)$

$= -\sin x + \cos x \neq f(x) \Rightarrow f(-x) \neq f(x)$

$\Rightarrow f(x)$ is neither even nor odd function

iii. $\sin^4 x + \cos^4 x$

Sol: $f(x) = \sin^4 x + \cos^4 x = (\sin x)^4 + (\cos x)^4$

$f(-x) = (\sin(-x))^4 + (\cos(-x))^4$

$= (-\sin x)^4 + (\cos x)^4 = \sin^4 x + \cos^4 x = f(x)$

$\Rightarrow f(-x) = f(x) \Rightarrow f(x)$ is an even function.

iv. $\tan x + \sec x$

Sol: $f(x) = \tan x + \sec x$

$f(-x) = \tan(-x) + \sec(-x)$

$= -\tan x + \sec x \neq f(x)$

$\Rightarrow f(-x) \neq f(x)$

$\Rightarrow f(x)$ is neither even nor odd.

CamScanner

v. $\dfrac{1}{\cos ec^3 x}$

Sol: $f(x)=\dfrac{1}{\cos ec^3 x}=\dfrac{1}{(\cos ecx)^3}$

$f(-x)=\dfrac{1}{(\cos ec(-x))^3}=\dfrac{1}{(-\cos ecx)^3}$

$=\dfrac{1}{-\cos ec^3 x}=-f(x)$

$\Rightarrow f(-x)=-f(x)\Rightarrow f(x)$ is an odd function.

vi. $\dfrac{\sin x+\sin 3x}{\cos x+\cos 3x}$

Sol: $f(x)=\dfrac{\sin x+\sin 3x}{\cos x+\cos 3x}$

$f(-x)=\dfrac{\sin(-x)+\sin 3(-x)}{\cos(-x)+\cos 3(-x)}=\dfrac{-\sin x-\sin 3x}{\cos x+\cos 3x}$

$=-\left(\dfrac{\sin x+\sin 3x}{\cos x+\cos 3x}\right)=-f(x)$

$\Rightarrow f(-x)=-f(x)$

$=f(x)$ is an odd function.

vii. $\dfrac{1}{\sec x+\sec^3 x}$

Sol: $f(x)=\dfrac{1}{\sec x+\sec^3 x}$

$f(-x)=\dfrac{1}{\sec(-x)+(\sec(-x))^3}$

$=\dfrac{1}{\sec x+(\sec x)^3}$

$=\dfrac{1}{\sec x+\sec^3 x}=f(x)$

$\Rightarrow f(-x)=f(x)\Rightarrow f(x)$ is an even function.

viii. $\dfrac{1}{\sec x+\cot^2 x}$

Sol: $f(x)=\dfrac{1}{\sec x+\cot^2 x}$

$f(-x)=\dfrac{1}{\sec(-x)+(\cot(-x))^2}$

$=\dfrac{1}{\sec x+(-\cot x)^2}=\dfrac{1}{\sec x+\cot^2 x}=f(x)$

$\Rightarrow f(-x)=f(x)$

$\Rightarrow f(x)$ is an even function.

Q2. Find the periods of the following functions.

i. $\sin 5x$

Sol: Since 2π is the period of $\sin x$ therefore

$\sin(5x+2\pi)=\sin 5x$

$\Rightarrow \sin 5\left(x+\dfrac{2\pi}{5}\right)=\sin 5x$

$\Rightarrow \dfrac{2\pi}{5}$ is the period of $\sin 5x$.

ii. $\cos 7x$

Sol: since 2π is the period of $\cos x$ therefore

$\cos(7x+2\pi)=\cos 7x$

$\Rightarrow \cos 7\left(x+\dfrac{2\pi}{7}\right)=\cos 7x$

$\Rightarrow \dfrac{2\pi}{7}$ is the period of $\cos 7x$.

iii. $\tan 3x$

Sol: Since π is the period of $\tan x$ therefore

$\tan(3x+\pi)=\tan 3x$

$\tan 3\left(x+\dfrac{\pi}{3}\right)=\tan 3x$

$\Rightarrow \dfrac{\pi}{3}$ is the period of $\tan 3x$.

iv. $\cot\dfrac{x}{2}$

Sol: Since π is the period of $\cot x$ therefore

$\cot\left(\dfrac{x}{2}+\pi\right)=\cot\dfrac{x}{2}$

$\Rightarrow \cot\dfrac{1}{2}(x+2\pi)=\cot\dfrac{x}{2}$

$\Rightarrow 2\pi$ is the period of $\cot\dfrac{x}{2}$.

v. $19\sin\left(\dfrac{\pi}{20}x\right)$

Sol: Since 2π is the period of $\sin x$ therefore

$19\sin\left(\dfrac{\pi x}{20}+2\pi\right)=19\sin\left(\dfrac{\pi x}{20}\right)$

$\Rightarrow 19\sin\dfrac{\pi}{20}(x+40)=19\sin\left(\dfrac{\pi x}{20}\right)$

$\Rightarrow 40$ is the period of $19\sin\left(\dfrac{\pi x}{20}\right)$

CamScanner

vi. $\mathbf{cosec\left(\frac{2x}{5}\right)}$

Sol: $cosec\left(\frac{2x}{5}\right)$

Since 2π is the period of $cosecx$ therefore

$$cosec\left(\frac{2x}{5}+2\pi\right)=cosec\frac{2x}{5}$$

$$\Rightarrow cosec\frac{2}{5}(x+5\pi)=cosec\frac{2x}{5}$$

$\Rightarrow 5\pi$ is the period of $cosec\frac{2x}{5}$

vii. $\mathbf{\frac{1}{2}sin\left(\frac{3x}{2}-\frac{\pi}{2}\right)}$

Sol: Period of $A\sin(Bx+C)+D=\frac{2\pi}{|B|}$

Period of $\frac{1}{2}\sin\left(\frac{3x}{2}-\frac{\pi}{2}\right)=\frac{2\pi}{3/2}=\frac{4\pi}{3}$

viii. $\mathbf{-5-3sec\left(7\pi x+\frac{\pi}{4}\right)}$

Sol: Period of $A\sec(Bx+C)+D=\frac{2\pi}{|B|}$

Period of $-5-3\sec\left(7\pi x+\frac{\pi}{4}\right)=\frac{2\pi}{7\pi}=\frac{2}{7}$

ix. $\mathbf{12+10\tan\left(\frac{\pi}{30}x\right)}$

Sol: Period of $A\tan(Bx+C)+D=\frac{\pi}{|B|}$

Period of $12+10\tan\left(\frac{\pi x}{30}\right)=\frac{\pi}{\pi/30}=30$

x. $\mathbf{6-4\cot\left(\frac{7x}{4}+\frac{\pi}{4}\right)}$

Sol: Period of $A\cot(Bx+C)+D=\frac{\pi}{|B|}$

Period of $6-4\cot\left(\frac{7x}{4}+\frac{\pi}{4}\right)=\frac{\pi}{7/4}=\frac{4\pi}{7}$

xi. $\mathbf{9+30sec\left(\frac{x}{15}+\frac{2\pi}{15}\right)}$

Sol: Period of $A\sec(Bx+C)+D=\frac{2\pi}{|B|}$

Period of $9+30\sec\left(\frac{x}{15}+\frac{2\pi}{15}\right)=\frac{2\pi}{1/15}=30\pi$

CamScanner